CS6140 - Machine Learning - Fall 2026
Final Project Proposal
What Is It Made Of? Predicting Clothing Material and Buyer Ratings from Amazon Product Listings

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II. Problem Description

Clothing material drives comfort, durability, and satisfaction, yet Amazon listings usually bury it in free-text bullets rather than a searchable field. We ask two questions: can a model infer a product's dominant material from its listing text, and does material, with price and department, help predict the product's average rating?

III. Data Summary

We use the Clothing, Shoes and Jewelry category of Amazon Reviews 2023 (Hou et al.): metadata for about 7.2 million products, each with a title, feature bullets, description, price, department, and average rating. We label each product by parsing composition strings such as "90% Polyester, 10% Spandex," taking the highest-percentage fiber and grouping rare materials into about ten classes.

IV. Methods

For material, we compare TF-IDF with logistic regression or a linear SVM against fine-tuned DistilBERT (Sanh et al.), with composition strings removed from the input so the task is not a lookup. For ratings, we compare random forest and XGBoost (Chen and Guestrin) regressors with a mean-rating baseline. We use five-fold cross-validation and report macro-F1 (classes are imbalanced) and RMSE/MAE. Challenges are noisy shoe labels, missing prices, and scale (we use a fixed-seed sample).

V. Preliminary Results

On the first 10,000 metadata records, only 3.6% of products store material in a structured field, but 46.0% give a percentage composition in their bullets and 79.0% name a known material. The top materials are polyester (24.3%), cotton (22.0%), leather (14.3%), and rubber (13.5%, mostly shoe soles). Price is missing for 45.7% of products.

VI. References

Chen, Tianqi, and Carlos Guestrin. "XGBoost: A Scalable Tree Boosting System." *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, ACM, 2016, pp. 785–94.

Hou, Yupeng, et al. "Bridging Language and Items for Retrieval and Recommendation." *arXiv*, 2024, arxiv.org/abs/2403.03952.

Sanh, Victor, et al. "DistilBERT, a Distilled Version of BERT: Smaller, Faster, Cheaper and Lighter." *arXiv*, 2019, arxiv.org/abs/1910.01108.

VII. AI Usage Statement

Claude (Anthropic) helped draft the text, write the sampling script, and format the document and MLA citations; the authors chose the questions, data, and methods and reviewed all text.